A Bi-Directional Dual-Path Boost–Buck Hybrid Converter for High-Voltage Power Transmission Delivery Cable in Humanoid Robots

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***Abstract*—Humanoid robots have great potential to replace human labors for various tasks in the near future. For a centralized energy storage system in humanoid robots, its battery pack (heart) is located in the main body while the energy is mostly consumed through its legs and hands, demanding thick wires (strong blood vessels) to deliver power to the actuators (muscles). In this work, we present a high-voltage power transmission cable system, which utilizes two bidirectional dual-path boost–buck (DP-BB) converters and reuses the parasitic cable inductor as the power inductor for a compact and efficient power delivery. The proposed power system reduces the cable current by 50%–67% and reduces the cable weight by 50%. In addition, the proposed phase alignment scheme can reduce the cable current ripple in such high-voltage conversion case. The converter was fabricated in a 0.18-μm BCD process, occupying an area of 3.56 $\times$ 5.55 mm^2 and a volume of 3.56 $\times$ 5.55 $\times$ 1.2 mm^3, including all the passive components. A 2-m 28 AWG power cable with a parasitic inductance of 1.99 μH and a resistance of 959 mΩ is utilized. It achieves a peak power conversion efficiency of 96.7% at 16.5-W output power and supports a maximum power delivery capacity of 45 W.**

***Index Terms*—48-V-bus power delivery, bidirectional charging, dc–dc converter, dual-path hybrid converter, high-voltage power transmission system, phase alignment.**

I. Introduction

HUMANOID robots have shown great potential to perform various dangerous, difficult and tedious tasks in the near future [1], [2], [3]. However, as these robots become more intelligent and multifunctional, their energy consumption rises significantly, making the operation duration a critical bottleneck for their large-scale deployment and broader application scenarios. Currently, the operation time of most humanoid robots is constrained to a couple of hours; the Atlas from Boston Dynamics, for example, can only operate for about 1 h. The main reasons behind this lie in limited battery volume, heavy weight, and low-system power efficiency.

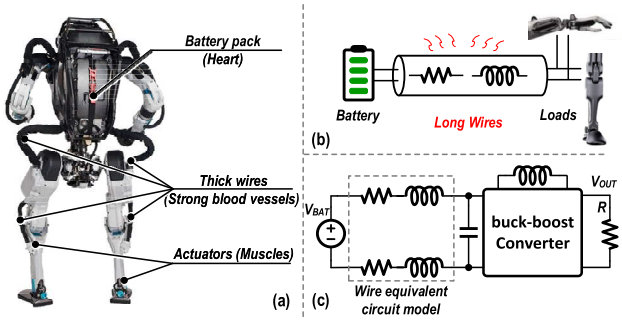


Fig. 1. (a) Humanoid robot with a centralized energy storage system. (b) Main power transmission path. (c) Corresponding conventional circuit implementation.

As illustrated in Fig. 1, a typical energy system architecture of humanoid robots relies on a centralized energy storage system, where the battery pack (heart) is located within the main body and supplies power to the actuators (muscles) which are distributed across the robot body and consume the majority of energy. In such a system, the energy loss mainly arises from two sources: power transmission loss and power conversion loss.

Power transmission losses originate from the parasitic resistance of cables, as energy must travel substantial distances from the battery to various load points via long transmission cables. This leads to significant I^2R losses. As shown in Fig. 2, one direct solution to mitigate the transmission losses is to use thicker transmission cables with lower resistance. However, thick cables occupy valuable space, complicating the integration of other functional components, and also contribute to an increase of the overall weight, which obviously degrade the energy efficiency and mobility. The conventional 12-V bus architecture has changed to 48-V bus architecture. The power distribution losses (I^2R losses) decrease by 16$\times$ with the quadrupling of the bus voltage. However, higher voltage (HV) batteries are heavier and complicate the design. Therefore, a boost-HV-buck system offers an effective solution to this issue.

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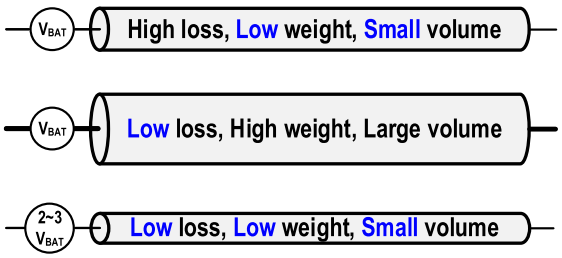


Fig. 2. Proposed high-voltage power delivery scheme and its benefits.

Power conversion losses stem from the losses inherent to the converters when regular voltage to meet different power supply requirements [4]. For the humanoid robots, the voltage gaps between the battery voltage and the specific voltage requirements of the actuators necessitate the use of dc–dc converters for voltage regulation. While these converters ensure proper voltage levels for actuator operation, they impose several design challenges in robot design. First, the inclusion of discrete bulky power inductors within these converters significantly impacts the overall space of the robot. Humanoid robots typically integrate dozens of actuators (e.g., for fingers, joints, and locomotion systems), each requiring an independent power conversion path to meet the specific voltage and current demands [5]. Furthermore, some precise components have very high requirements for volume. For example, coreless motors used in robot fingers and joints typically weigh between 5 and 50 g, with a diameter ranging from 10 to 30 mm and a length not exceeding 40 mm. If the inductors take up too much space, it will affect the flexibility. Moreover, these converters introduce additional losses during operation and generate excess heat, complicating the thermal management and reducing overall system efficiency. As the power demand increases, the physical space required for dc–dc converters and the associated heat both grow proportionally.

Recent advancements in dc–dc converter designs have been focused on switched-capacitor and inductor hybrid converter designs to increase the power conversion efficiency and power density [6], [7], [8]. One notable example is the inductor-first buck (IFB) converter [9], which places the inductor at the low-current input terminal to reduce inductor current and thus improves the conversion efficiency. As shown in Fig. 3, the IFB converter also allows the reutilization of the cable parasitic inductor as a power inductor to eliminate the bulky inductor in conventional dc–dc converters and to significantly increase the power density [10], [11], [12], [13], [14]. In addition, another work in [15] proposes a voltage regulating cable (VRC) which integrates the charging IC and its passives onto the USB Type-C connectors, thereby moving the charger converter and its associated heat out of the device. However, there is only one chip actively performing the voltage regulation while the other chip is by-passed, leading to non-optimized system efficiency [16].

In this article, we propose a boost–buck power transmission cable with a bi-directional dual-path hybrid converter for a power-efficient energy system in humanoid robots [17]. The hybrid converter on the battery (transmitter) side boosts the voltage on the cable, reducing the current and transmission losses. Then, the converter on the load (receiver) side steps the voltage down to the required level to power the actuators. In addition, a dual-path hybrid converter was implemented to achieve the desired boost and buck voltage conversion by reusing the cable inductor as the power inductor. This design not only achieves voltage regulation but also considerably reduces power transmission losses and conversion losses. Furthermore, we propose a phase alignment scheme to reduce the cable current ripple in HV power conversion cases. In addition, the operational endurance of humanoid robots is inherently constrained by battery capacity; any effective energy recovery mechanism is helpful for extending their operational range. The proposed bidirectional power transmission can also be utilized to recycle power to the battery. Humanoid robots perform dynamic motions involving frequent acceleration and deceleration of limbs. During deceleration or passive movements, the actuators in the limbs operate in regenerative mode, converting kinetic energy back into electrical energy. Notably, leading robotics developers have incorporated energy recovery systems in their robot designs (e.g., Tesla Optimus and Boston Dynamics Atlas) as a proven approach to enhance endurance.

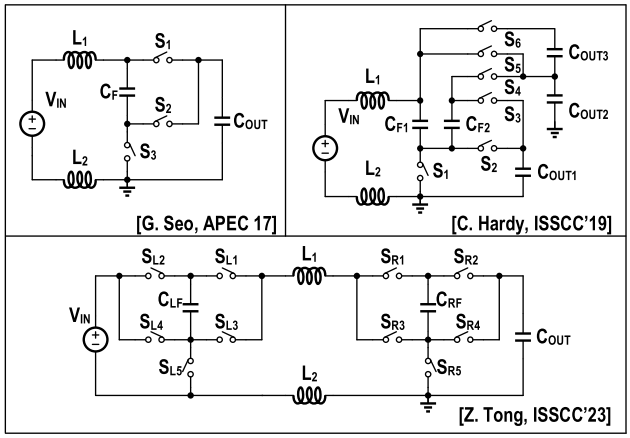


Fig. 3. Prior arts utilizing the USB cable as the power inductor.

The organization of this article is as follows. Section II presents the cable loss analysis and introduces operation principle of the proposed boost-HV-buck power transmission cable. Section III gives the system block diagram and the circuit implementation of the cable emulator and the reconfigurable bootstrap capacitor circuit. Section IV exhibits the measurement results of the high-voltage transmission cable. Finally, Section V draws the conclusion of this work.

II. Proposed HV Transmission System

A. Proposed High-Voltage Power Transmission Solution

As illustrated in Fig. 4, a buck–boost converter is typically employed to bridge the voltage gap between the battery and the load requirements. Traditionally, this converter is located on the load side to ensure a better output voltage regulation [18], [19], [20], [21], [22], [23]. In this work, we split the conventional buck–boost converter into a boost converter on the battery side and a buck converter on the load side,

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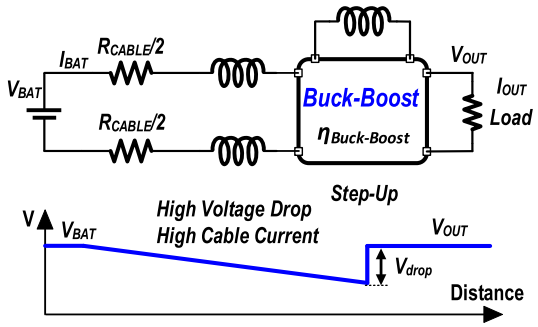


Fig. 4. Conventional buck–boost solution in battery-powered robots.

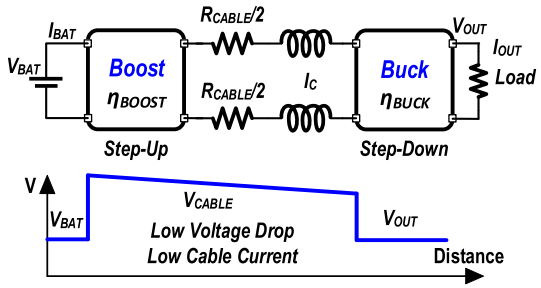


Fig. 5. Proposed boost-HV-buck solution for battery-powered robots.

enabling HV transmission through the cable, as depicted in Fig. 5. Compared with the traditional approach, increasing the voltage level on the cable helps reduce the I^2R transmission losses. Taking the converter efficiency into account, the end-to-end efficiency of the conventional and proposed transmission systems is analyzed below.

For conventional direct transmission, the input power can be expressed as the sum of the output power and the total losses. The total losses include the cable loss and the chip loss. Therefore, the input power can be written as follows:

$$P_{\mathrm{IN}} = P_{\mathrm{OUT}} + P_{\mathrm{Chip_loss}} + P_{\mathrm{Cable_loss}} \tag{1}$$

where $P_{\mathrm{Chip_loss}}$ and $P_{\mathrm{Cable_loss}}$ represent the loss on the buck–boost chip and the transmission cable, respectively, and the losses and cable current can be expressed as follows:

$$\begin{cases} P_{\mathrm{Chip_loss}} = \left(\dfrac{1}{\eta_{\mathrm{Buck\text{-}Boost}}} - 1\right) P_{\mathrm{OUT}} \\ P_{\mathrm{Cable_loss}} = I_{\mathrm{Cable}}^2 R_{\mathrm{Cable}} \\ I_{\mathrm{Cable}} = \dfrac{M_{\mathrm{Buck\text{-}Boost}}}{\eta_{\mathrm{Buck\text{-}Boost}}} I_{\mathrm{OUT}} \end{cases} \tag{2}$$

where $\eta_{\mathrm{Buck\text{-}Boost}}$, $M_{\mathrm{Buck\text{-}Boost}}$, and R_{Cable} represent the efficiency of load-side buck–boost chip, the buck–boost chip voltage conversion ratio (VCR), and equivalent resistance of the transmission cable, respectively. In this situation, by solving equations above, the system efficiency can be expressed as follows:

$$\eta_D = \frac{P_{\mathrm{OUT}}}{P_{\mathrm{IN}}} = \frac{\eta_{\mathrm{Buck\text{-}Boost}} P_{\mathrm{OUT}}}{P_{\mathrm{OUT}} + \frac{M_{\mathrm{Buck\text{-}Boost}}^2}{\eta_{\mathrm{Buck\text{-}Boost}}} I_{\mathrm{OUT}}^2 R_{\mathrm{Cable}}}. \tag{3}$$

For the proposed high-voltage transmission case, the input power can also be expressed as the sum of the output power and the total losses. The total losses include the cable loss and the losses of two chips. Therefore, the input power can be written as follows:

$$P_{\mathrm{IN}} = P_{\mathrm{OUT}} + P_{\mathrm{Boost_loss}} + P_{\mathrm{Buck_loss}} + P_{\mathrm{Cable_loss}} \tag{4}$$

where $P_{\mathrm{Boost_loss}}$, $P_{\mathrm{Buck_loss}}$, and $P_{\mathrm{Cable_loss}}$ represent the loss on the boost chip, the buck chip, and the transmission cable, respectively, and the losses and cable current can be expressed as follows:

$$\begin{cases} P_{\mathrm{Boost_loss}} = \left(\dfrac{1}{\eta_{\mathrm{Boost}}} - 1\right) P_{\mathrm{OUT}} \\ P_{\mathrm{Boost_loss}} = \left(\dfrac{1}{\eta_{\mathrm{Buck}}} - 1\right) P_{\mathrm{OUT}} \\ P_{\mathrm{Cable_loss}} = I_{\mathrm{Cable}}^2 R_{\mathrm{Cable}} \\ I_{\mathrm{Cable}} = \dfrac{M_{\mathrm{Buck\text{-}Boost}}}{\eta_{\mathrm{Buck\text{-}Boost}}} I_{\mathrm{OUT}} = \dfrac{\eta_{\mathrm{Buck}}}{M_{\mathrm{Buck}}} I_{\mathrm{BAT}} \end{cases} \tag{5}$$

where η_{Buck}, η_{Boost}, M_{Buck}, M_{Boost}, and R_{Cable} represent the efficiency of the load-side buck chip, the efficiency of the battery-side boost chip, VCR of the buck chip, VCR of the boost chip, and the equivalent resistance of the transmission cable, respectively. In this situation, the system efficiency can be calculated as follows:

$$\eta_{\mathrm{HV}} = \frac{P_{\mathrm{OUT}}}{P_{\mathrm{IN}}} = \frac{\eta_{\mathrm{Boost}} \eta_{\mathrm{Buck}} P_{\mathrm{OUT}}}{P_{\mathrm{OUT}} + \frac{M_{\mathrm{Buck}}^2}{\eta_{\mathrm{Buck}}} I_{\mathrm{OUT}}^2 R_{\mathrm{Cable}}}. \tag{6}$$

Although the actual performance would be influenced by multiple factors, we can still qualitatively compare the two (3)–(6). Specifically, HV transmission systems demonstrate superior performance, where larger cable resistance is present. This is because HV levels reduce the required current, thereby minimizing resistive losses in the transmission cables. However, large conversion ratios in both boost and buck converters can compromise the converter's conversion efficiency, highlighting a tradeoff between transmission efficiency and power conversion efficiency. In the application of humanoid robots, the advantages of high-voltage transmission become more pronounced due to the necessity of long transmission paths from a centralized battery pack to distributed actuators located in the limbs. Consequently, the reduction in cable losses offered by HV transmission is beneficial in such applications.

Moreover, Table I presents a quantitative comparison of three scenarios that utilize two different cables [as shown in Fig. 6(a)] of different gauges and distinct transmission architectures. All configurations are designed to deliver an output of 18 V at 3 A, assuming that all the dc–dc conversions have an efficiency of 97% for all converters. When comparing different transmission methods using the same cable, the proposed HV transmission approach achieves a 15% improvement in E2E efficiency, which is particularly beneficial for humanoid robot applications. Furthermore, due to its higher efficiency, the HV transmission scheme can operate with thinner cables, saving space for functional components within the robot.

B. Parasitic Analysis of the Transmission Cable

In addition, the power transmission cable used in this work is shown in Fig. 6(b). The electrical characteristics of the transmission cable can be modeled using two wires, one power

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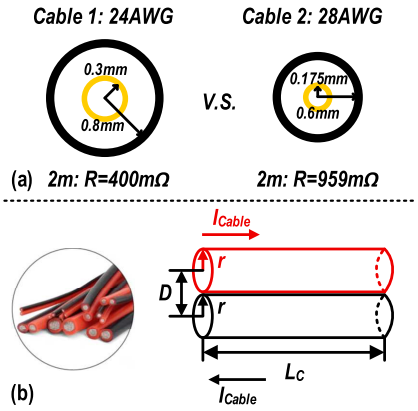


Fig. 6. (a) Cross-sectional view and comparison of 24 AWG and 28 AWG cables. (b) Photograph of the power transmission cable used in this work.

TABLE I
COMPARISON OF TWO POWER SYSTEMS

Case	Transmission Method	Transmission Cable	Transmission Efficiency
1	HV Transmission	2m Cable 2	91.7%
2	Conventional Transmission	2m Cable 2	76.8%
3	Conventional Transmission	2m Cable 1	89.8%

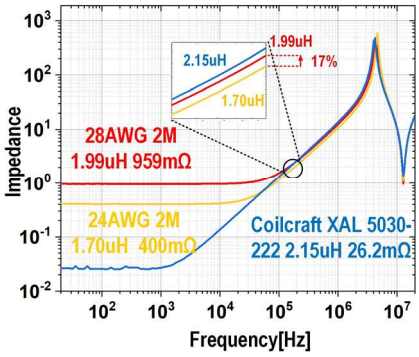


Fig. 7. Measured transmission cable impedance with different thicknesses.

wire and one ground wire. Distance D of the two wires is a constant. The inductance of the cable can be written as follows:

$$L = \frac{\mu_0 L_C}{\pi}\left(\ln\frac{D}{r} + \frac{1}{4}\right) \tag{7}$$

where L_C represents the cable length and r denotes the wire radius. The magnetic constant, μ_0, which defines the strength of a magnetic field, is given as $4\pi \times 10^{-7}$. As shown in our previous work [16], the shape of the cable has no effect on the inductance due to the constant distance D between the power wire and the ground wire.

Moreover, thinner cables exhibit higher parasitic inductance, making them advantageous for performing as power inductors. Fig. 7 presents the measured impedance of cables with different thicknesses, compared with the Coilcraft XAL5030-222 inductor. The plots show that, the cables behave similar to the power inductor with a self-resonant frequency around 5–10 MHz. Also, the inductance of a 2-m cable is close to 2 μH, where the 28 AWG cable demonstrates higher inductance than

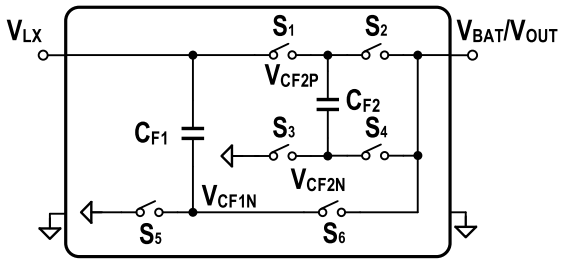


Fig. 8. Proposed bi-directional dual-path boost-HV-buck hybrid converter.

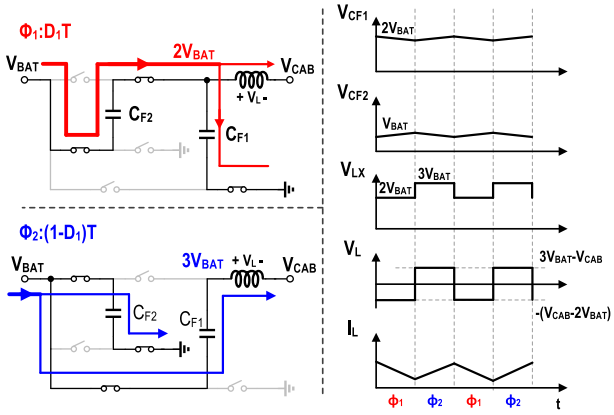


Fig. 9. Boost mode operation phase and waveforms.

the 24 AWG one, better supporting the cable reutilization and HV power transmission.

C. Proposed Dual-Path Boost–Buck (DP-BB) Converter

Fig. 8 presents the proposed DP-BB hybrid converter architecture, which consists of two flying capacitors (C_{F1} and C_{F2}) and six switches (S_1–S_6).

Fig. 9 depicts the operation phase of DP-BB chip when configured in boost mode. In this mode, the converter operates in two phases. During φ_1, (DT) switches S_1, S_4, and S_5 are on, while S_2, S_3, and S_6 remain OFF. The battery source V_{BAT} provides energy to V_{CBL} and C_{F1} through C_{F2}, while the cable inductor L de-energize. In φ_2 with $(1-D)T$ duration, switches S_2, S_3, and S_6 are ON, while S_1, S_4 and S_5 are OFF. The input voltage V_{BAT} charges C_{F2} and delivers energy to the cable through C_{F1}, while the cable inductor L remains in an energizing state. By applying the principle of inductor volt-second balance, the VCR and cable inductor current in boost mode can be derived as follows:

$$\frac{V_{CAB}}{V_{BAT}} = 3 - D_1 (2 < M < 3) \tag{8}$$

$$I_L = \frac{1}{3 - D_1} I_{BAT}. \tag{9}$$

Conversely, the buck mode operation can be achieved by reversing the input and output terminals of the converter operating in boost mode, as illustrated in Fig. 10. The operating phases in buck mode closely resemble those of the boost mode. By applying the principle of inductor volt–second balance, the

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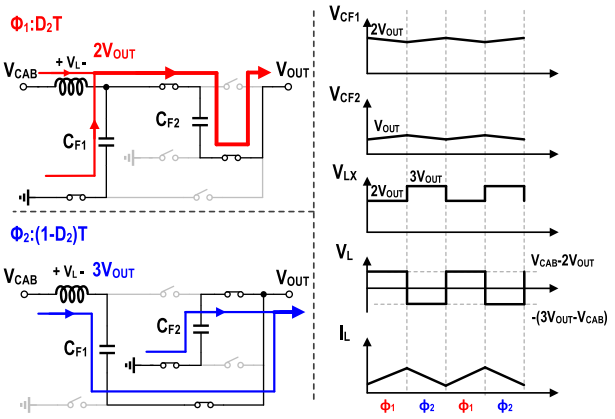


Fig. 10. Buck mode operation phase and waveforms.

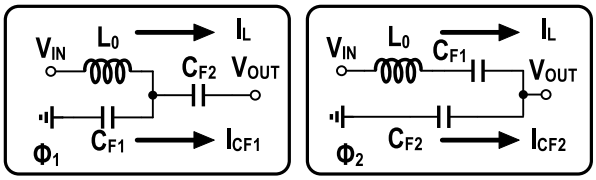


Fig. 11. Operational phases of the proposed dual-path converter.

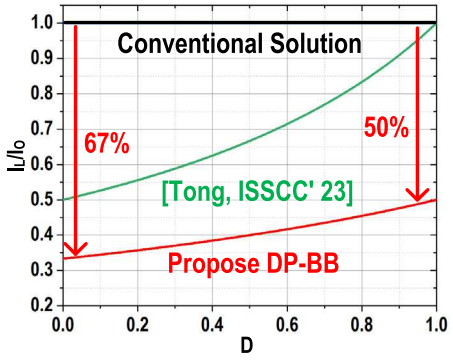


Fig. 12. Normalized cable current compared with prior works.

VCR and inductor current in the buck mode can be expressed as follows:

$$\frac{V_{\mathrm{OUT}}}{V_{\mathrm{CAB}}} = \frac{1}{3-D_2}\left(\frac{1}{3} < M < \frac{1}{2}\right) \tag{10}$$

$$I_L = \frac{1}{3-D_2} I_{\mathrm{OUT}}. \tag{11}$$

As illustrated in Fig. 11, an additional capacitor current path (I_{CF1}, I_{CF2}) is in parallel with the inductor, delivering a portion of the output current and reducing the current through the inductor (I_L) [24], [25]. As demonstrated in Fig. 12, proposed DP-BB architecture achieves a 50%–67% I_L reduction across the full duty cycle (D) range compared with the conventional solution.

For a dc–dc converter, the power loss of switches can be expressed as follows:

$$P_{\mathrm{LOSS}} = 2\sqrt{R_{\mathrm{ON},0}E_{\mathrm{SW},0}}\sqrt{f_{\mathrm{SW}}}\sum_{i=1}^{N_S} I_{\mathrm{RMS},i}V_{\mathrm{DS},i} \tag{12}$$

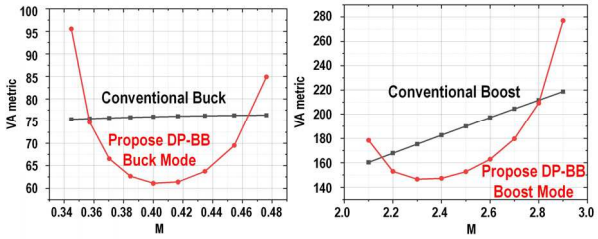


Fig. 13. Normalized VA metric compared with conventional solutions.

TABLE II
PARAMETERS OF POWER SWITCHES

	Switch	VBLOCK	Duty	ID/IO
Buck	S1	VIN	D	1
	S2	VIN	1-D	1
Proposed Converter Buck Mode	S1	2VOUT	D	1/((3-D)D)
	S2	VOUT	1-D	1/((3-D)(1-D))
	S3	VOUT	1-D	1/((3-D)(1-D))
	S4	VOUT	D	1/((3-D)D)
	S5	VOUT	D	(1-D)/((3-D)D)
	S6	VOUT	1-D	1/(3-D)
Boost	S1	VOUT	D	1/(1-D)
	S2	VOUT	1-D	1/(1-D)
Proposed Converter Boost Mode	S1	2VIN	D	1/D
	S2	VIN	1-D	1/(1-D)
	S3	VIN	1-D	1/(1-D)
	S4	VIN	D	1/D
	S5	VIN	D	(1-D)/D
	S6	VIN	1-D	1

where $R_{\mathrm{ON},0}$ and $E_{\mathrm{SW},0}$ are the baseline device metric values for conduction resistance and switching energy, respectively. f_{SW} is the operation switching frequency. $V_{\mathrm{DS},i}$ is the drain–source voltage of switches, and $I_{\mathrm{RMS},i}$ is the equivalent root-mean-square (rms) current flowing through the switches. The minimum power losses in one switch directly depends on $V_{\mathrm{DS},i}$ and $I_{\mathrm{RMS},i}$, which is VA metric. Table II shows the V_{DS}, I_D, and duty cycle of different switches of the proposed converter and conventional buck and boost converters. The total VA metric comparison is shown in Fig. 13. It shows that our converter has a better performance in both buck mode and boost mode.

For HV transmission, the converter incorporates a boost mode on the source side to step up the battery voltage and a buck-mode chip on the load side to regulate the output, enabling bidirectional charging and seamless mode transitions. This bidirectional capability allows identical chips to be deployed at both ends of the power transmission cable, with each chip dynamically regulated to implement bidirectional battery charging. Then, the energy can be recycled back to the battery pack through the proposed bidirectional power conversion.

D. HV Transmission Mode With Phase Alignment

Fig. 14 illustrates the proposed boost-HV-buck converter architecture with two DP-BB chips and the equivalent cable inductors. The two chips operate cooperatively: the transmitter chip boosts the voltage on the transmission cable, while the receiver chip performs output voltage regulation. In this configuration, both terminals of the cable inductor are switching

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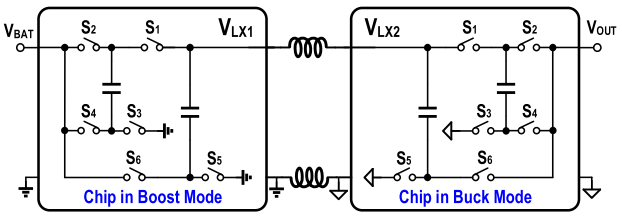


Fig. 14. Proposed boost-HV-buck converter architecture with two DP-BB chips and the equivalent cable inductors.

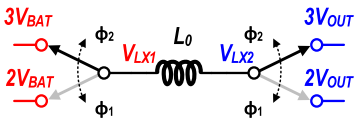


Fig. 15. Simplified switching node voltage levels of the HV transmission mode.

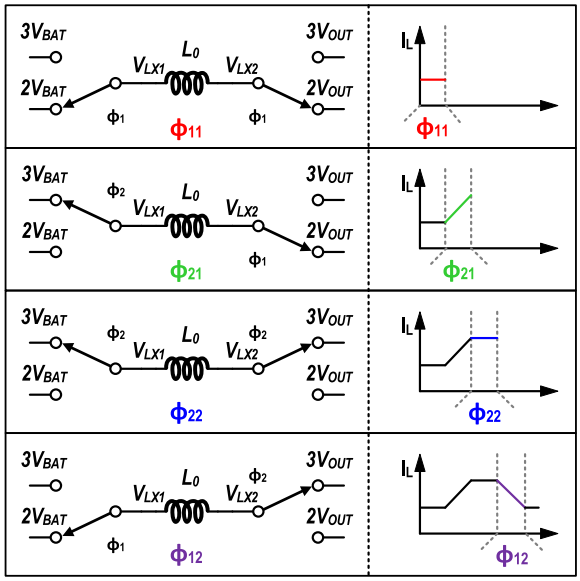


Fig. 16. Operation phase of high-voltage transmission mode.

nodes. Specifically, as illustrated in Fig. 15, the left-side switching node V_{LX1} alternates between $2V_{BAT}$ and $3V_{BAT}$, while the right-side node V_{LX2} switches between $2V_{OUT}$ and $3V_{OUT}$.

As illustrated in Fig. 16, four possible switching phase combinations between the two chips result in distinct voltage conditions across the cable inductor L_0, each influencing the inductor current differently.

In phase φ_{11}, both chips operate in φ_1. The switching nodes are $V_{LX1} = 2V_{BAT}$ and $V_{LX2} = 2V_{OUT}$, resulting in an inductor voltage of $V_L = 2V_{BAT} - 2V_{OUT}$. When $V_{BAT} \approx V_{OUT}$, the inductor current ripple remains nearly constant.

In phase φ_{21}, chip 1 operates in φ_2, while chip 2 operates in φ_1. The switching nodes are $V_{LX1} = 3V_{BAT}$ and $V_{LX2} = 2V_{OUT}$, yielding $V_L = 3V_{BAT} - 2V_{OUT}$. In this case, L_0 is energized, and the inductor current increases significantly.

In phase φ_{22}, both chips operate in φ_2. The switching nodes are $V_{LX1} = 3V_{BAT}$ and $V_{LX2} = 3V_{OUT}$ resulting in

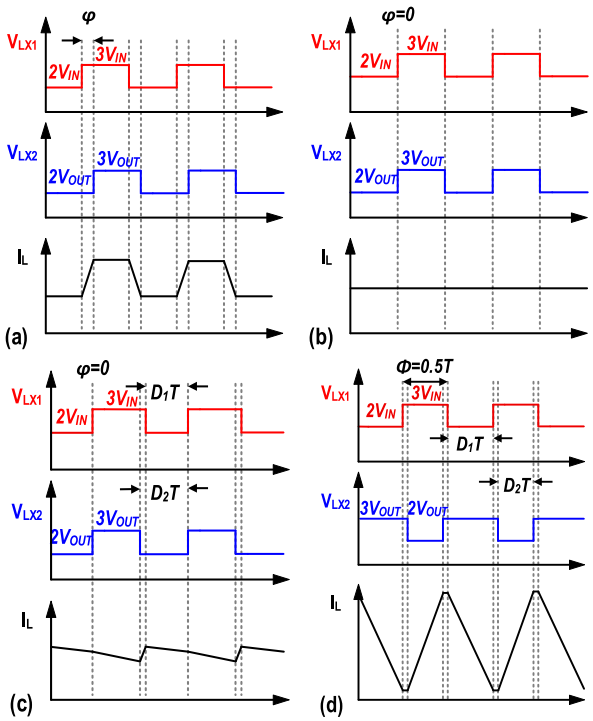


Fig. 17. (a) Operation phase of the proposed phase alignment for hysteretic control, (b) theoretical best case of phase alignment, (c) actual best case of phase alignment, and (d) worst case of phase misalignment.

$V_L = 3V_{BAT} - 3V_{OUT}$. Again, when $V_{BAT} \approx V_{OUT}$, the inductor current remains approximately constant.

In phase φ_{12}, chip 1 operates in φ_1 and chip 2 in φ_2. The switching nodes are $V_{LX1} = 2V_{BAT}$ and $V_{LX2} = 3V_{OUT}$, yielding $V_L = 2V_{BAT} - 3V_{OUT}$. In this condition, L_0 is de-energized, and the inductor current decreases significantly.

By applying the steady-state equation, the VCR of the transmission mode can be expressed as follows:

$$M = \frac{V_{OUT}}{V_{BAT}} = \frac{3 - D_1}{3 - D_2} (0.67 < M < 1.5) \tag{13}$$

$$I_L = \frac{1}{3 - D_2} I_{OUT}. \tag{14}$$

The proposed boost-HV-buck converter achieves an E2E VCR ranging from 0.67 to 1.5, effectively functioning as a boost–buck converter. As discussed previously, the cable current ripple is governed by the voltage difference between the boost-side switching node V_{LX1} and the buck-side switching node V_{LX2}. As illustrated in Fig. 16, the cable current ripple increases when the durations of the non-overlapping phases (φ_{12} and φ_{21}) are extended while the overlapping phases (φ_{11} and φ_{22}) are shortened. Conversely, extending the overlapping phases and reducing the duration of non-overlapping phases lead to lower current ripple.

E. Analysis of the Phase Mismatch and Current Ripple

The analysis of the relationship between the phase mismatch and the cable current ripple is shown as follows.

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The waveforms of V_{LX1}, V_{LX2}, and I_L are shown in Fig. 17(a). For the inductor current can be written as follows:

$$v = L\frac{di}{dt}. \tag{15}$$

Assume $V_{IN} = V_{OUT} = V$, the difference of two chip working phase is φ. The cable current ripple can be written as follows:

$$\Delta i = \frac{V\phi}{L}. \tag{16}$$

For the best case of phase alignment, that is, when $\varphi = 0$, the two chips align their phases as close as possible, as shown in Fig. 17(b). We get the smallest cable current ripple and the smallest loss. However, considering the voltage drop on the power MOSFET, the actual voltage of V_{LX1} is smaller than $2V_{IN}$ and $3V_{IN}$, the actual voltage of V_{LX2} is larger than $2V_{OUT}$ and $3V_{OUT}$. Therefore, the actual best case phase alignment is shown in Fig. 17(c). The two chips duty cycles are not the same. D_2T is larger than D_1T. The cable current ripple can also be written as follows:

$$\Delta i = \frac{v(D_1 - D_2)T}{L}. \tag{17}$$

For the worst case for the two chips, as shown in Fig. 17(d), the two chips operate with full mismatch when $\varphi = 0.5$ T. We get the largest cable current ripple and the largest loss. The cable current ripple can be given as follows:

$$\Delta i = \frac{vD_1T}{L} = \frac{vD_2T}{L}. \tag{18}$$

Consequently, when the phase difference of the two chips reaches 0, the cable current ripple is the smallest. To minimize cable current ripple and to improve overall efficiency, phase alignment between the two chips is critical. However, achieving direct communication between the two converter chips using analog methods is challenging due to the long physical distance of the power transmission cable, while digital methods require additional hardware blocks and data transmission cable. This separation imposes significant constraints on direct duty-cycle synchronization and necessitates a complex analog/digital interface and clock synchronization system.

To address this challenge, we propose an analog hysteretic control scheme on the buck-side chip to track the operation of the boost-side chip without requiring long-distance signal transmission. Details of this control strategy are presented in Section III.

III. Circuit Implementations

A. System Block Diagram

The system block diagram of the proposed DP-BB converter is presented in Fig. 18. The circuit enclosed within the gray dotted box represents the on-chip implementation, which is divided into two functional blocks: the power stage and the control stage. The chip integrates all power switches, the associated bootstrap circuitry, and the controller. Off-chip components include two flying capacitors C_{F1} and C_{F2}, and an external current-sensing resistor R_{SEN} used for measuring the current on the transmission cable.

In the power stage, the architecture incorporates six high-voltage lateral-diffused NMOS (LDNMOS) transistors (S_1–S_6), each paired with its own gate driver. To maintain the required bootstrap voltage during operation, active diodes (D_1, D_2, D_4, and D_6) are employed, working in conjunction with on-chip bootstrap capacitors (C_{b1}, C_{b2}, C_{b4}, and C_{b6}).

In the control stage, two analog multiplexers (MUXs) are used for mode selection among three operation modes: buck mode, boost mode, and hysteretic control mode. The standalone buck and boost modes are used for single-chip functional testing. The hysteretic control mode is activated on the load side to synchronize with the source-side chip, enabling inter-chip phase alignment and reducing cable current ripple.

For voltage-mode control in typical dc–dc converters, the output voltage is fed back to the controller for regulation. However, in thr boost mode, direct feedback is challenging due to the switching behavior at the V_{LX} node. To overcome this, a cable emulator is employed to estimate the average voltage across the transmission cable [16], enabling effective regulation. In addition, two key circuit techniques are implemented in this work to enable reduced inductor current ripple and reduced on-chip bootstrap capacitor size: 1) hysteretic control for phase alignment and 2) an optimized bootstrap circuit design. These techniques are detailed in Sections III-B and III-C.

B. Hysteretic Control for Phase Alignment

To minimize the cable current ripple, a hysteretic control scheme for inter-chip phase alignment is proposed, as illustrated in the system block diagram in Fig. 19. In this configuration, the left-side converter operates in the boost mode using a primary pulsewidth modulation (PWM)-based voltage control loop to autonomously generate its switching signals. In contrast, the right-side converter, operating in the buck mode, utilizes secondary hysteretic current control, which is synchronized through local current sensing.

A current-sensing resistor R_{SEN} is placed at the ground terminal of the buck-side chip rather than the high-voltage side. This strategic placement eliminates the need for complex high-voltage signal processing and improves the current sensing accuracy. In this work, the sensing resistance is 10 mΩ. Compared with the cable resistance of 959 mΩ, it is only 1% of the cable resistance. Given the relatively high resistance of the long transmission cable, the addition of R_{SEN} introduces only negligible efficiency loss. The sensed current ripple voltage V_{SEN} is amplified and compared with a control error signal via a hysteresis comparator, which then generates the phase control signal for the right-side converter. The voltage corresponding to the sensed cable current can be expressed as follows:

$$V_{SEN} = -I_L R_{SEN} \tag{19}$$

$$\overline{V_{SEN}} = -\frac{V_{SEN}R_2}{R_1}. \tag{20}$$

The hysteretic phase-alignment control operation is depicted in Fig. 19 and comprises four operational phases.

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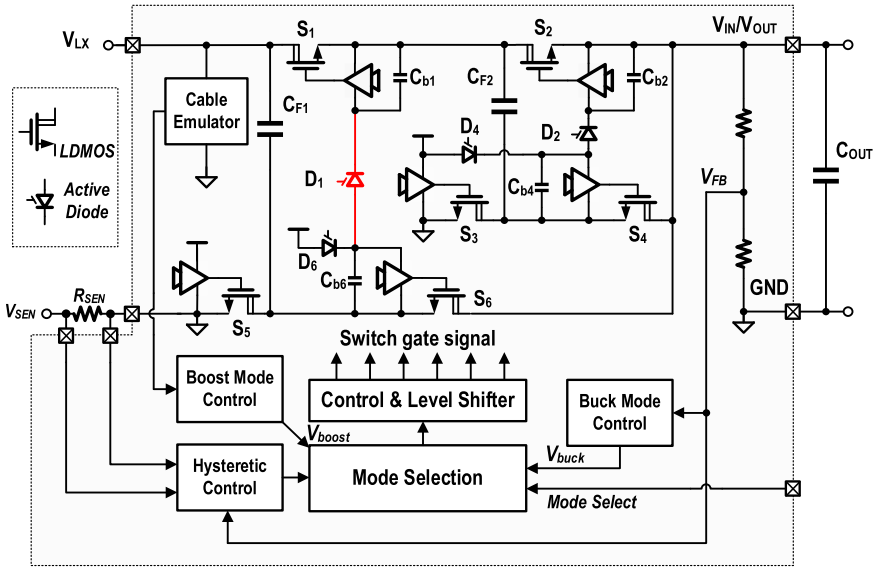


Fig. 18. System block diagram of the proposed DP-BB converter.

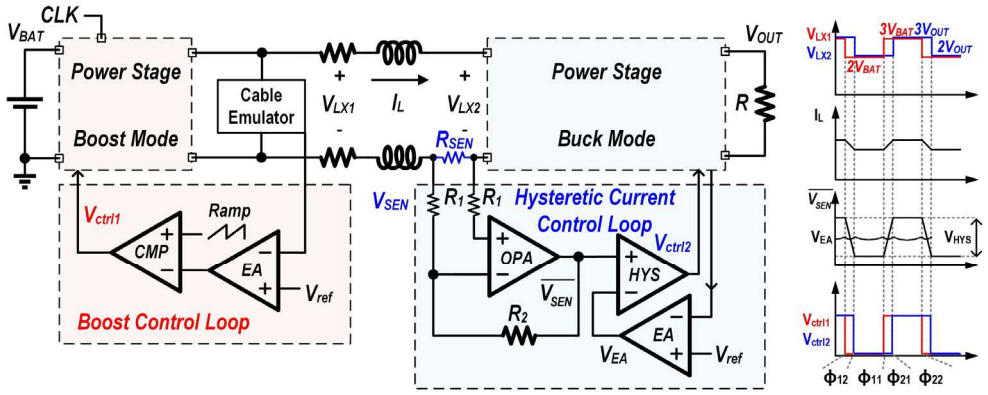


Fig. 19. Proposed hysteretic control for phase alignment.

1) φ_{12}: The cable current decreases significantly until the sensed current signal intersects the lower threshold of the hysteresis window. At this point, the right-side chip transitions from φ_2 to φ_1, initiating φ_{11}.
2) φ_{11}: The cable current remains nearly constant, maintaining alignment between the two chips.
3) φ_{21}: Triggered by the left-side chip switching from φ_1 to φ_2, the cable current increases sharply. When the upper hysteresis boundary is crossed, the right-side chip enters φ_{22}.
4) φ_{22}: The cable current stabilizes again, and the cycle repeats when the left-side chip returns to φ_1, re-entering φ_{12}.

This hysteretic control confines the inductor current within a defined window, thereby suppressing excessive current ripple and improving the transmission efficiency. Although it is not feasible to completely eliminate the non-overlapping phases φ_{21} and φ_{12}, careful design of the hysteresis window and amplifier gain can significantly reduce the current ripple and mitigate the worst-case effects of phase misalignment between the two chips. In addition, shorter or longer cables led to changes in the inductance value of the cable. For traditional converters, this will lead to differences in cable current ripple. However, in hysteresis control, the cable current ripple does not change with the variation of cable length, or it can be said that the ripple is only slightly affected due to the digital control delays.

Fig. 20(a) illustrates the operation phase of the proposed phase alignment for hysteretic control. When we reduce the size of the hysteresis window, the voltage signal of the cable current will also arrive at the hysteresis window in advance, as depicted in Fig. 20(b); when we increase the gain of the amplifier, the voltage signal of the cable current will reach the hysteresis window earlier, as shown in Fig. 20(c). In both cases, the cable current ripple will be smaller. However, an excessively large amplifier gain or an overly small hysteresis window may lead to false triggering of the hysteresis window.

Conversely, when we increase the size of the hysteresis window, the voltage signal of the cable current will also reach the hysteresis window later, as illustrated in Fig. 20(d); when we decrease the amplifier gain, the voltage

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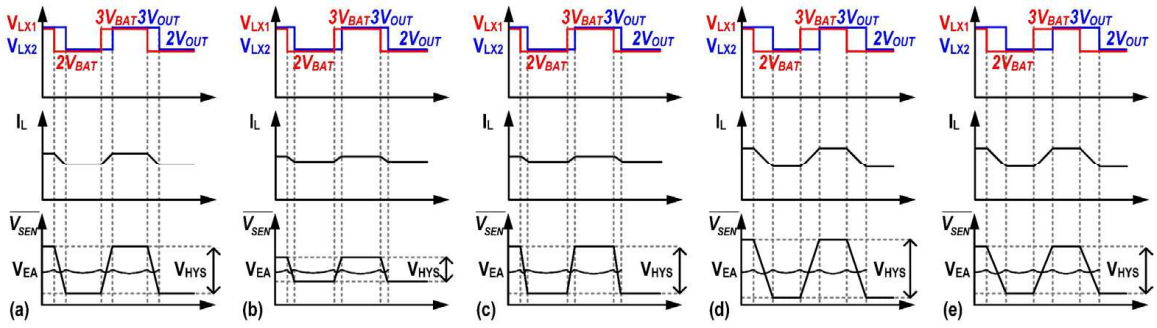


Fig. 20. Operation phase of the proposed phase alignment for hysteretic control using (a) proposed gain of amplifier and hysteresis window, (b) smaller hysteresis window, (c) higher gain of amplifier, (d) larger hysteresis window, and (e) smaller gain of amplifier.

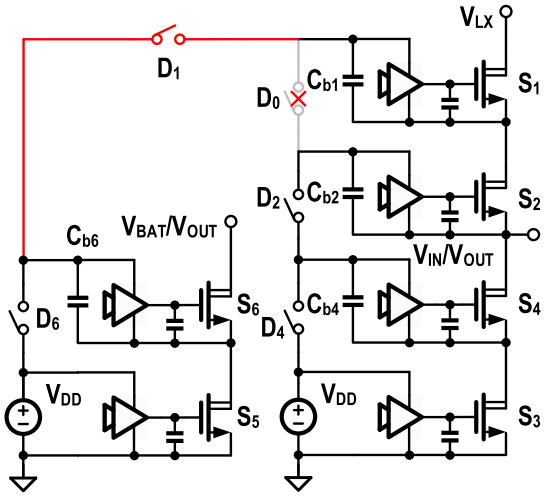


Fig. 21. Bootstrap circuit implementation with traditional bootstrap method and proposed capacitance-efficient bootstrap circuit.

signal of the cable current will reach the hysteresis window later, as shown in Fig. 20(e). These two scenarios will result in larger cable current ripple and, consequently, increased losses.

Therefore, the optimal hysteresis window size and amplifier gain should be determined based on the conditions that neither cause false triggering of the hysteresis window nor increase the phase mismatch. In this work, the hysteresis window size is 400 mV and the amplifier gain is 40 dB.

C. Proposed Bootstrap Circuit Implementation

As illustrated in Fig. 21, cascaded bootstrap approaches are commonly employed in circuits with multiple floating switches [26], [27]. These methods utilize the bootstrap capacitor of a lower side switch to charge that of the next higher side switch in the stack. However, as the number of cascaded power switches increase, the required bootstrap capacitance for the low-side switches becomes significantly larger. Given the relatively low capacitance density in CMOS fabrication processes, such large capacitors occupy substantial silicon area. Therefore, optimizing the required capacitance while maintaining sufficient gate drive voltage is critical in achieving compact integration.

TABLE III
CALCULATED BOOTSTRAP CAPACITANCE

Conventional Bootstrap		Proposed Bootstrap	
	Capacitance		Capacitance
C_{b1}	$30C_G$	C_{b1}	$20C_G$
C_{b2}	$60C_G$	C_{b2}	$20C_G$
C_{b4}	$90C_G$	C_{b4}	$40C_G$
C_{b6}	$10C_G$	C_{b6}	$40C_G$
C_{total}	$190C_G$	C_{total}	$120C_G$

In this work, we analyze the conventional cascaded bootstrap method under three ideal assumptions: 1) a minimum gate drive voltage of $0.9 \times V_{DD}$ is necessary to ensure low ON-resistance of the switches; 2) uniform voltage redistribution value of all bootstrap capacitors; and 3) each power switch has the same gate capacitance denoted by C_G. With these assumptions, the total required bootstrap capacitance of conventional bootstrap implementations reaches approximately $190C_G$, with the lower side bootstrap capacitor C_{b4} accounting for $90 \times C_G$, dominating area consumption.

To address this challenge, we propose a capacitance-efficient bootstrap circuit that is also shown in Fig. 21. Instead of charging C_{b1} from C_{b2}, the proposed method utilizes C_{b6} to charge C_{b1}. This is enabled by the simultaneous switching of transistors S_6 and S_2, which ensures the source nodes of S_6 and S_1 are at the same potential. The calculated bootstrap capacitance is shown in Table III. With the same assumption, this technique reduces the required bootstrap capacitance to $120C_G$, thereby enabling area-efficient integration without sacrificing gate-drive performances.

IV. MEASUREMENT RESULTS

The proposed DP-BB converter was fabricated using a 0.18-μm BCD process and occupies a chip area of 2.8×2.4 mm^2, as shown in Fig. 22. The pad arrays for V_{LX}, V_{C1N}, V_{C2P}, and V_{C2N} are placed on chip to facilitate direct soldering of two 10-μF flying capacitors (C_{F1} and C_{F2}, 0603 package). Two additional 22-μF 0603 capacitors serve as the output capacitor C_{OUT}. The converter operates at a switching frequency of 1 MHz and occupies a PCB area of 3.56×5.55 mm^2 and

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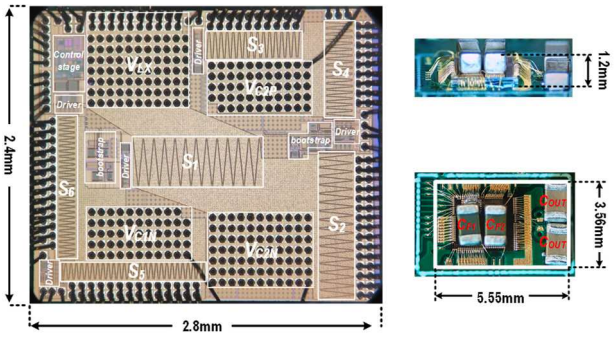


Fig. 22. Chip micrograph of the proposed bidirectional DP converter.

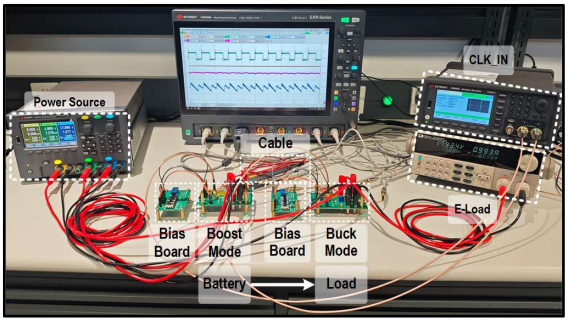


Fig. 23. Test environment of high-voltage transmission mode.

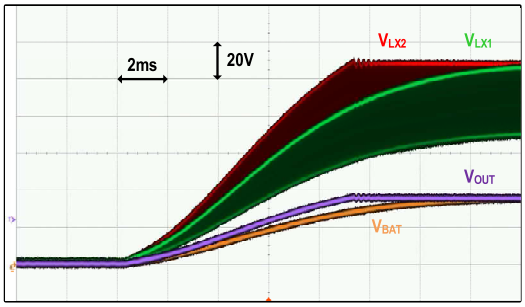


Fig. 24. Measured start-up waveform of 18–48–18 V.

a volume of 3.56 × 5.55 × 1.2 mm^3. Notably, the design eliminates the need for on-board passive power inductors; instead, a 2-m-long 28 AWG transmission cable functions as the power inductor, enabling a highly compact solution.

Fig. 23 illustrates the experimental setup for high-voltage power transmission, emulating a long-distance battery-to-load power delivery scenario in a humanoid robotic system. A programmable voltage source supplies the input power, while an electronic load serves as dynamic load. Two DP-BB chips are placed at the two ends of the 2-m transmission cable. The left-side chip operates in boost mode to step up the input voltage, while the right-side chip functions in the buck mode to regulate the output voltage.

Fig. 24 presents the start-up behavior of the high-voltage transmission system. As the source voltage ramps up to 18 V, the output voltage is regulated to 18 V within 14 ms. The switching nodes, V_{LX1} and V_{LX2}, swing between $2V_{BAT}/3V_{BAT}$ and $2V_{OUT}/3V_{OUT}$, respectively, aligning with the operating principles.

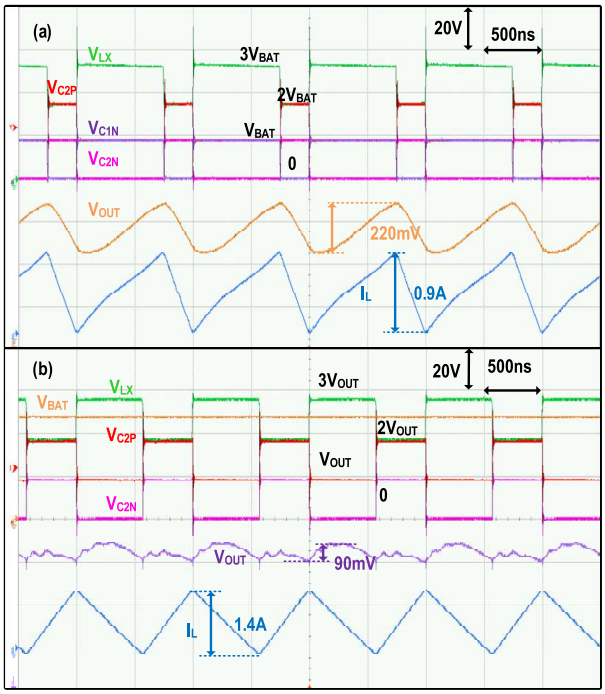


Fig. 25. Measured steady-state waveforms of single chip (a) boost mode for the 18–48-V conversation and (b) buck mode for the 48–18-V conversation.

Fig. 25(a) shows the steady-state waveforms for the single-chip work in boost mode. The output voltage shows the average voltage of the transmission cable V_{LX}. The test waveform shows the voltage stepping from 18 up to 48 V at an output current of I_{OUT} = 0.5 A, achieving an output voltage ripple of 220-mV and 0.9-A cable current ripple. Fig. 25(b) presents the steady-state waveforms in buck mode. The input voltage is the average voltage of the transmission cable. The test waveform shows a 48–18-V down conversion at I_{OUT} = 1.5 A, resulting in a voltage ripple of 90 mV and a cable current ripple of 1.4 A. In this work, all measurement results are obtained with a cable of 2-m 28 AWG.

Fig. 26 shows the comparison of the steady-state performance of phase-misaligned and phase-aligned dual-chip operation at V_{BAT} = 18 V, V_{CAB} = 48 V, and V_{OUT} = 18 V. In the worst-case phase misalignment scenario, the cable current ripple reaches 2.5 A. In contrast, optimal phase alignment through hysteretic control reduces the ripple to 0.5 A, achieving an 80% reduction. This significant improvement highlights the effectiveness of the proposed phase-alignment strategy in minimizing ripple and enhancing overall system efficiency.

Fig. 27 shows the measured results of hysteretic control in transmission mode, demonstrating the effectiveness of phase alignment scheme. The system exhibits a hysteresis window of 400 mV and an output signal delay of 20 ns. The voltage sensing signal V_{SEN}, representing the inverted polarity of the voltage across the sensing resistance, is processed through an inverting amplifier. The resultant waveform $\overline{V_{SEN}}$ directly corresponds to the cable current I_L.

Fig. 28 quantifies the efficiency performance of the proposed HV the proposed HV solution maintains stable efficiency above 90% in heavy load conditions, in contrast to

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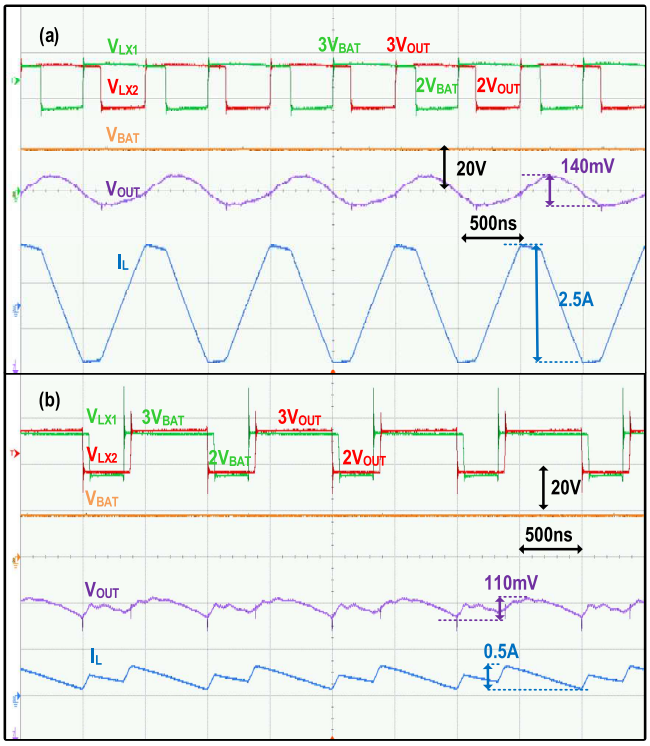


Fig. 26. Measured steady-state waveforms of transmission mode under the conditions V_{BAT} = V_{OUT} = 18 V when scenario of (a) phase misalignment and (b) phase alignment.

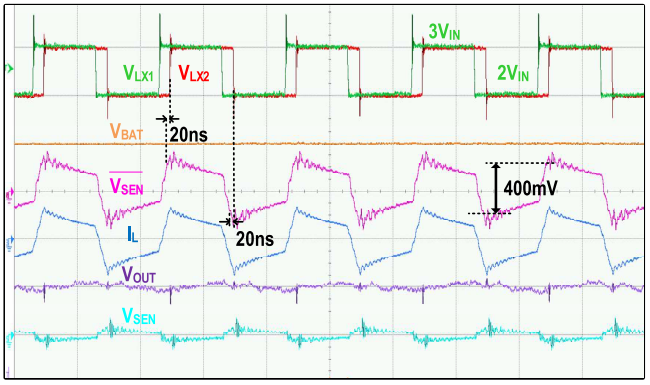


Fig. 27. Measured steady-state waveforms of hysteretic control in the transmission mode.

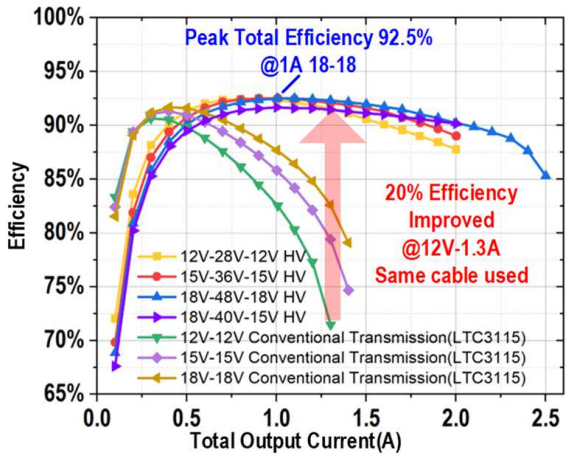


Fig. 28. Measured efficiency of system efficiency.

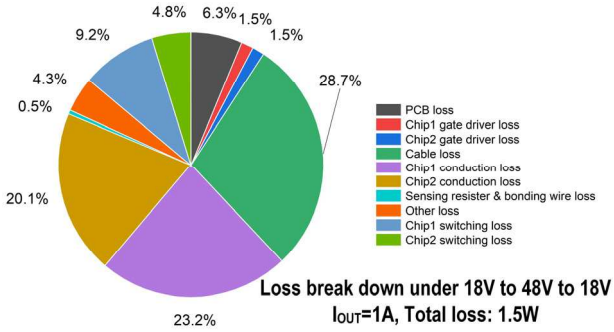


Fig. 29. Power loss breakdown for the case of 18–48–18 V.

traditional implementations that suffer from substantial voltage degradation due to resistive cable losses. For the coreless motors, the input voltage range of these motors is 3.7–24 V, with 12-V systems being commonly used. Their required rated current is 0.1–1 A. The proposed power supply system is capable of powering coreless motors and satisfying their operational needs. During light load, although the light load chip efficiency is lower than heavy load, the total power is smaller. In order to improve the light load efficiency, the converter can work in discontinuous conduction mode (DCM) and pulse-frequency modulation (PFM) and group some of the power MOSFETs for adaptive sizing during light load in the future works. During heavy load, the 2.5-A measurement is the maximum output current of the proposed converter in this work, not the upper limit of the converter. In face of different power requirements, the size of power transistors can be increased to meet the requirement. Meanwhile, the cables should also be replaced with thicker ones to match this change. When thicker cables are adopted, their parasitic inductance will decrease; however, at the same time, the current ripple allowed under heavy-load conditions can be larger, which will not affect the stability of the entire power supply system.

Fig. 29 shows the pie chart in the case of 18–48–18-V conversion when I_{OUT} = 1 A. The conduction losses (chip 1 + chip 2) account for 43.4% (23.2% + 20.14%), constituting the largest share due to steady-state current flow in the power devices. The cable transmission losses constitute 28.8%, which are significantly reduced by the high conversion ratio (as shown in Fig. 12). The switching losses combined make up 13.9% for chip 1 and chip 2 in total. While the peak currents do contribute to these losses, their portions remain relatively low due to optimized gate driving and device selection, mitigating the impact of hard switching at high conversion ratios. The sensing resister together with bonding wire only accounts 0.5% losses during the power transmission.

Fig. 30 presents the single-chip efficiency of both buck and boost operating modes. The efficiency curves with the cable loss show the test system efficiency, it contains the chips and cable. The efficiency curves without cable are estimated by subtracting the I_{AVG}^2R cable loss, where I_{AVG} is

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TABLE IV
PERFORMANCE SUMMARY AND COMPARISON WITH PRIOR ARTS

	ISSCC'19[13]	TIE'20[11]	ISSCC'23[15]		ISSCC'23[18]	LTC3115	This work		
Converter In Cable	NO	NO	YES		NO	NO	**YES**		
Inductor Type	Cable	Cable	Cable		Discrete	Discrete	**Cable**		
Used Cable	USB 1m	USB 1m	USB 2m	USB 1.5m	/	/	**Low Weight 28AWG Cable 2m**		
Conversion Direction	Unidirectional	Unidirectional	Bidirectional		Bidirectional	Unidirectional	**Bidirectional**		
Conversion Type	Step-Down	Step-Down	Forward Mode	Reverse Mode	Step-Up &Step-Down	Step-Up &Step-Down	**Buck Mode**	**Boost Mode**	**Transmission Mode**
			Step-Down	Step-Down			**Step-Down**	**Step-Up**	**Step-Up& Step-Down**
Input Voltage[V]	9	5	12-20	8-12.6	5-48	2.7-40	**28-48**	**12-18**	**12-18**
Output Voltage[V]	6-8	3.3-3.9	8-12.6	5-9	5-48	2.7-40	**12-18**	**28-48**	**12-18**
Capacitor[F]	C_1:2*10μ +2*3.3μ C_2:1*10μ +2*3.3μ	C_{F1}:2*22μ C_{F2}:4*10μ C_{O1}:2*22μ C_{O2}:4*22μ C_{O3}:1*22μ	C_F:1*10μ C_L:1*10μ		8*(C_F:10μ) C_O:4.7μ	C_O	**C_{F1}:10μ C_{F2}:10μ C_O:2*22μ**		**2*(C_{F1}:10μ) 2*(C_{F2}:10μ) C_O:2*22μ**
$I_{L,DC}/I_{OUT}$ Range	1/2~1	1/3~1/2;1/2~1	1/2~1	1	2~3	1	**1/3~1/2**	**1**	**1/3~1/2**
Inductor Location	Input	Input	Input	Output	Middle	Middle	**Input**	**Output**	**Middle**
Peak Output Power[W]	31.9	15.2	42	18	15	48	**54**	**43.2**	**45**
Switching Frequency[MHz]	1.8-2.3	2	5	5	2	1	**1**	**1**	**1**
Peak Efficiency	93.5%	89.7%	92.8%	92.4%	/	/	**95.8%**	**94.1%**	**92.5%**
Peak Chip Efficiency	97.6%	N.A.	95.1%	93.7%	95.4%@Buck 93.2%@Boost	96.0%	**96.9%**	**95.2%**	**93.8%**
Power Density @PE [W/mm²]	0.32	0.15	1.90	0.41	N.A.	N.A.	**0.99**	**0.85**	**0.83**
Peak Power Density[W/mm²]	0.69	0.28	3.17	1.36	N.A.	N.A.	**2.73**	**2.19**	**1.13**

*Calculated value without cable loss,**Estimated value according to paper or datasheet,***High E2E efficiency due to phase alignment

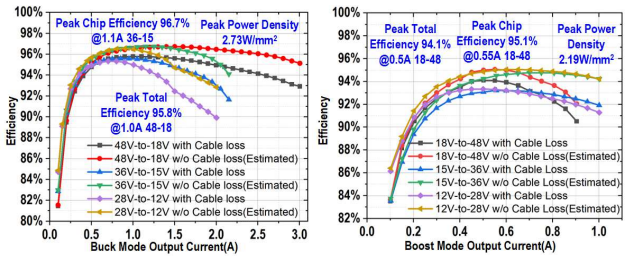


Fig. 30. Measured efficiency of (a) buck mode chip efficiency and (b) boost mode chip efficiency.

the average cable current. The actual chip efficiency is higher than the estimation because of the cable current ripple. In buck mode [see Fig. 30(a)], a peak efficiency of 96.7% is achieved (excluding the cable loss) during 36–15-V conversion at 1.1-A output current, with a corresponding power density of 2.73 W/mm^2 at 54-W output power. In the boost mode [see Fig. 30(b)], the converter reaches 95.1% peak efficiency during 18–48-V conversion at 0.55-A output current, achieving a maximum power density of 2.19 W/mm^2 at 43.2-W output power.

Table IV summarizes and compares the performance of the proposed DP-BB converter with state-of-the-art designs. To the best of the authors' knowledge, this work presents the first implementation of a dual-chip architecture that collaborates with a transmission cable to support bidirectional HV power transmission. The proposed solution achieves the highest reported peak output power while significantly reducing cable current. In addition, the proposed solution offers a substantial reduction in system weight without compromising efficiency, making it well-suited for humanoid robots.

V. CONCLUSION

This work presents a bidirectional DP-BB hybrid converter that repurposes the transmission cable as a power inductor while employing dual-path operation and high-voltage transmission mode to reduce cable current by 50%–67%. The topology enables bidirectional voltage conversion with a 20% efficiency improvement over conventional solution. Measurement results show that in the buck mode, the peak power density is 2.73 W/mm^2 at 54-W output power, with 96.5% peak efficiency. In the boost mode, the peak power density is 2.19 W/mm^2 at 43.2-W output, with 95.1% peak efficiency. The proposed boost–buck scheme may achieve higher system efficiency gain with dedicated power transmitter (boost) and power receiver (buck) chips.

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